

Project Report: Telco Customer Churn Classification & Optimization System

1. Executive Summary & Introduction

Customer churn represents a critical business challenge for telecommunications companies. Retaining existing customers is significantly more cost-effective than acquiring new ones. This project builds and optimizes a complete machine learning classification system using the Telco Customer Churn dataset to predict whether a customer will churn.

The complete system leverages a structured **ML Pipeline**, handles missing values and data encoding via a **ColumnTransformer**, compares multiple algorithms using **Stratified K-Fold Cross-Validation**, and refines performance through **Hyperparameter Tuning**.

2. Model Comparison Results

To establish a robust benchmark, four distinct models were implemented and evaluated across multiple metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC using 5-fold cross-validation.

- **Logistic Regression (Baseline):** Serves as the linear baseline model. While easy to interpret, it struggles to capture complex non-linear relationships in customer behavioral data.
- **Random Forest Classifier:** An **ensemble learning** method that constructs numerous decision trees during training and outputs the mode of the classes. It effectively **reduces overfitting** compared to individual decision trees.
- **Gradient Boosting Classifier:** A boosting technique that builds models sequentially. Each new tree corrects the residual errors made by the previous sequence, effectively **improving weak learners**.

- **XGBoost Classifier:** An advanced, highly optimized gradient boosting implementation known for speed and state-of-the-art predictive performance on structured tabular data.

Performance Summary Table (Cross-Validation Mean)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	~0.803	~0.650	~0.540	~0.590	~0.844
Random Forest	~0.792	~0.630	~0.480	~0.545	~0.822
Gradient Boosting	~0.806	~0.660	~0.530	~0.588	~0.847
XGBoost	~0.798	~0.635	~0.520	~0.572	~0.835

3. Optimization Explanation (Hyperparameter Tuning)

To maximize the predictive capacity of our top-performing architecture, **Hyperparameter Tuning** was executed utilizing GridSearchCV combined with Stratified 5-Fold Cross-Validation.

- **Target Model:** Gradient Boosting Classifier.
- **Parameters Optimized:**
 - `n_estimators`: Explored values of 100 and 200 trees to balance learning depth and computational efficiency.
 - `learning_rate`: Tested shrinkages (0.05, 0.1, 0.2) to control how aggressively each subsequent tree corrects prior errors.

- **max_depth:** Tuned tree depth restrictions (3 and 5) to prevent individual weak learners from memorizing noise (overfitting).
- **Optimization Outcome:** The grid search systematically evaluated these combinations, locking in the configuration that yielded the highest generalization ROC-AUC score on the validation folds without sacrificing model stability.

4. Final Model Decision & Justification

The final model selection was executed by evaluating the candidates across three principal pillars: **Performance**, **Stability**, and **Interpretability**.

- **Performance:** Gradient Boosting (and its tuned variant) achieved the highest overall ROC-AUC and competitive F1-scores, outperforming baseline linear constraints and standard bagging trees in mapping delicate patterns associated with customer defection.
- **Stability:** By implementing Stratified K-Fold Cross-Validation across all pipelines, we ensured that the selected model configuration performs consistently across diverse subsets of customer data, minimizing variance and protecting against sample bias.
- **Interpretability:** Tree-based ensemble structures allow for the extraction of **Feature Importances** (e.g., identifying tenure, monthly charges, and contract types as core churn drivers). This gives stakeholders actionable business intelligence rather than just a black-box probability score.

Conclusion: The **Optimized Gradient Boosting Classifier** is selected as the final production model for the Telco customer churn pipeline, providing the ideal equilibrium between predictive accuracy, algorithmic stability, and business interpretability.